

**"Studies on root rot disease of chickpea
(*Cicer arietinum* L.) caused by *Macrophomina phaseolina*
with special reference to its integrated management"**

THESIS

Submitted to the

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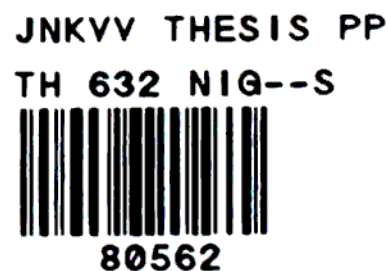
**In partial fulfilment of the requirements for
the Degree of**

MASTER OF SCIENCE

In

**AGRICULTURE
(PLANT PATHOLOGY)**

By



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2004

*Dedicated to
my beloved Parents*

CERTIFICATE - I

*This is to certify that the thesis entitled "Studies on root rot disease on chickpea (*Cicer arietinum* L.) caused by *Macrophomina phaseolina* with special reference to its integrated management " submitted in partial fulfilment of the requirement for the degree of **MASTER OF SCIENCE IN AGRICULTURE** of the Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, is a record of the bonafied research work carried out by **Shri Om Prakash Nigodia** under my guidance and supervision. The subject of the thesis has been approved by the student's Advisory Committee and the Director of Instructions.*

No part of this thesis has been submitted for any other degree or diploma (certificate awarded etc.) or has been published/published part has been fully acknowledged. All the assistance and help received during the course of the investigation has been acknowledged by him.

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CERTIFICATE -II

*This is to certify that the thesis entitled "Studies on root rot disease on chickpea (*Cicer arietinum* L.) caused by *Macrophomina phaseolina* with special reference to its integrated management " submitted by **Shri Om Prakash Nigodia** to Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, in partial fulfilment of the requirement for the degree of M.Sc. (Ag.) in **DEPARTMENT OF PLANT PATHOLOGY** has been after evaluation, approved by the External Examiners and by the students Advisory Committee after an oral examination on the same.*

Place : Sehore (M.P.)

Date : 10/3/05

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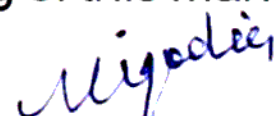
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Om Prakash Nigodia

INTRODUCTION

CHAPTER - I

Introduction

The chickpea (*Cicer arietinum* L.) is also known as gram, bengal gram and chana in Hindi. It belongs to the family leguminosae. It is the major pulse crop used in diet of vegetarians in India and is a good source of protein. It contains 21.1 g protein, 4.5 g fat, 150 mg minerals, 61.5 g of carbohydrates and some malic and oxalic acid per 100 g seed. Besides these other important minor elements of great nutritive values are also contained in it. Dal and Besan (flour) are the important forms in which it is consumed by people. Malic and oxalic acids collected from green leaves are prescribed for intestinal disorders. Chickpea is the most important pulse accounting for more than one third of the area and about 40% production of pulses in India. Chickpea is generally grown as a rainfed crop in the Rabi season sometimes it is also grown as a normal or partially irrigated crop. It is best suited to areas having low to moderate rainfall and mild cold weather. In India it is grown in area of 63.05 lakh hectares with a production of 53.90 lakh tonnes. (Anonymous, 2002). In Madhya Pradesh chickpea is cultivated throughout the state particularly in Bhopal, Vidisha, Sehore, Gwalior, Bhind, Morena, Shivpuri, Guna, Jabalpur, Narsinghpur, Sagar, Damoh, and Chhindwara Districts. It occupies 2553.2 thousand ha with an annual production 240.1 thousand tonnes in the state (Anonymous, 2002). The average yield of chickpea in M. P. comes to 908 kg/ha which is quite low in comparison to international standards. A number of reasons may be assigned for low yield of chickpea out of them the general tendency of farmers to grow the pulses the energy rich crop on poor and eroded soil and negligence in the adoption of plant protection measures against diseases and pests are the main. Chickpea is the victim of a large number of diseases and insect pests. The major yield losses are caused by diseases occurring from pre-emergence to the maturity of crop. Most of the diseases of chickpea are caused by fungi associated with seed and present in soil. Since, the diseases are mainly soil borne hence it can be best controlled by using resistant varieties. However, it has been observed that a variety showing resistance at one place exhibits susceptible reaction at

other place. Out of these root rot disease caused by *Macrophomina phaseolina* (*Rhizoctonia batiticola*) is of wide spread occurrence causing serious losses and is one of the limiting factors in increasing its production. Climatic conditions play a predominant role in determining the course and severity of the disease epidemic. The information on this aspect of *Macrophomina phaseolina* of chickpea is very sparse. Basically, it is soil borne disease and difficult to manage through any single method hence integration of all available approaches like biological, chemical and intercrops is required to manage the disease.

The present investigation was undertaken with the following objectives:

1. To survey and record the frequency occurrence of root rot disease of chickpea in and around Sehore.
2. To study the incidence of root rot disease in relation of meteorological parameters, edaphic factors and host genotype.
3. To evaluate the efficacy of various seed dresser against *Macrophomina phaseolina* *in vitro* and *in vivo*.
4. Evaluation of bio agents and their substrate for the management of root rot of chickpea.
5. To find out the integrated disease management practices.

REVIEW OF LITERATURE

CHAPTER - II

Review of Literature

The first report of occurrence of root rot in chickpea along with wilt was made by Padwick (1948). It was the worst disease of chickpea with complex nature. *Fusarium*, *Operculella padwickii* and other non parasitic causes were later reported with this disease. Kaiser *et al.* (1968) observed *Macrophomina phaseolina* associated in periodic surveys on wilt and root rot incidence of gram.

Gupta *et al.* (1983) reported incidence of root rot ranging 3.58 to 20.63 per cent in 30 villages of Northern Madhya Pradesh. Reddy *et al.* (1990) observed 100 per cent mortality due to *Rhizoctonia bataticola* (*M. phaseolina*) in JG 62 variety of chickpea. This is one of the most common diseases in South-East Asian countries and limits chickpea production (ICRISAT Annual Report 1986; FAO 1990).

In other countries like Mandalay, Sagaing and Bago, central and lower Myanmar the major disease in chickpea was dry root rot caused by *R. bataticola* (Reddy *et al.* 1991). Aung *et al.* (1992) also observed that dry root rot caused by *R. bataticola* (*M. phaseolina*) was the major disease in these countries. In Ethiopia root rot disease of chickpea was also observed by Beniwal *et al.* (1992). Significant reduction (25-70%) in chickpea production is caused by this disease in India (Ahmad *et al.* 1986; Pandey and Singh, 1990 and Ahmad and Mouselli, 1990).

Pathogenicity :

Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne diseases generally appearing at two phases of growth (i) early seedling stage and (ii) flowering stage. Wilting was due to *F. oxysporum* f. sp. *ciceri*, root rot caused by *Rhizoctonia bataticola* and *Sclerotium rolfsii* resulted in collar rot.

Byadgi and Hegde (1985) studied the variations in size and shape of pycnidial and pycnidiospores of *M. phaseolina* on chickpea based on morphological cultural characters and pathogenic behaviour. Than et al. (1991) found dry root rot causing isolates of *M. phaseolina*. The isolates in soyabean and *Phaseolus vulgaris* and chickpea were related. Demirci et al. (1999) isolated *M. phaseolina* from plant and seed of chickpea cultivar Azizive - 94 and tested pathogenicity.

Epidemiology :

Dry root rot caused by *Rhizoctonia bataticola* is a serious disease whenever the crop is exposed to temp. more than 30°C (Singh and Mehrotra 1982). The disease development was influenced by dry soil condition especially at flowering. The plant suddenly dries in the field. The leaves and stem of affected plant are usually straw coloured. The tap root is dark and quite brittle in dry soil. The dark black sclerotia can be observed on and within host tissue. *R. bataticola* is soil borne and sclerotia, formed on the organic residue and in the host tissue, become the chief source of inoculum. The pathogen has a wide host range.

Management (Cultural practices) :

Singh (1976) found significant inhibition in growth of *Rhizoctonia bataticola* with the extract of soil amended with immature and mature chickpea crop. Dhrub et al. (1987) observed gradual decrease in the population of *M. phaseolina* and significantly lower infection percentages in wheat and oat amended soil.

Taya et al. (1988) observed that incidence of *R. bataticola* (*M. phaseolina*) on sandy than on clay soil and low soil moisture were conducive to disease development. Disease was enhanced by increased level of nitrogen (N) fertilizer and decreased by increased Phosphorus (P). When both fertilizers were added together the effect of N. was masked by that of Phosphorus (P).

Rush and winter (1990) observed that no significant differences were found when chickpea followed the other crops either previous crops also affected residual soil $\text{No}_3\text{-N}$ in general residual soil cotton fallow or sunflower plots but the differences were not always significant.

Smiley *et al.* (1992) observed that root rot in chickpea was not as prevalent when glyphosate was applied 1 or 2 daysy direct drilling compared with applications made 3 days before when tillage or application of glyphosate was performed in the autumn or in spring.

Gupta (2000) observed that chickpea is attacked by root pathogen *R. bataticola* where is most destructive in causing root rot. Micronutrients have an important role in disease resistance. Laboratory studies showed that out of the 6 micro-nutrients tested, zinc inhibited the growth of the pathogen.

Biological control :

The use of fungal antagonists like *T. viride* and *T. harzianum* was found effective against root rot of chickpea caused by *M. phaseolina* (Saxena and Saxena, 1995).

Singh *et al.* (1998) also used *Trichoderma viride* and *T. harzianum* as seed treatment, soil inoculum and soil inoculum + seed treatment proved effective in reducing the incidence of dry root rot *fungus. R. bataticola (M.phaseolina)* by increasing the phenolics and carbohydrates contents of chickpea. *Pseudo monas fluorences. Bacillus subtilis* and *Aspergillus flavus* were also effective.

Rangashwaran and Prasad (2000) targeted *F. oxysporum* f. sp. *ciceri. R. solani* and *M. phaseolina* (root rot pathogens) by four selected antagonists under green house condition. The maximum plant stand (100%) was observed with *P. fluorescens* (PDBCAB 29 and 30) treated pots for *R. solani* and *M. phaseolina*, *P. putida* (PDBCAB 19) and *P. fluorescens* (PDBCAB 30) were able to fully control the disease.

Shaid and Shrivastava (2000) conducted trials on evaluation of the activity of various plant products (0.1% palmarosa oil, 0-3 % neem oil, 10% *Ocimum tenuiflorum* leaf extract) along with various antagonistic micro organism (*Bacillus subtilis* log cfu/ml) *Pseudomonas fluorecen* (500 g/hac), *Trichoderma viride* (10g cfu/ml) *Aspergillus flavus* (10g/ml) against dry root rot (*R. bataticola*) in chickpea under pot conditions. Various seed treatments, soil inoculations and seed treatment plus soil inoculations combination were assessed. All treatments gave some control. Palmarosa oil gave the best control approximate 22-32 per cent disease incidence compared with 66 per cent in control and soil inoculation plus seed treatment was the most effective application method.

Chemical control :

Verma, (1976) found seed treatment with bavistin or carboxin at the rate of 0.25 per cent to protect seedling in field for 30 days or more from root rot of chickpea.

Taya *et al.* (1990) observed that the clay content of the soil reduced the efficacy of carbendazin alone or in combination applied as a seed treatment or soil drench. Increased levels of N reduced and p level enhanced the effectivity of carbendazim and carbendazim + thiram applied as seed treatment or pre-sowing drenching.

Mathieson and Rush (1991) observed that seed treatment with triadimenol had significantly greater emergence and dry matter production in chickpea in root rot infested field.

Rajinder et al. (1998) found that seed treatment + soil drenching was the most effective method of application in reducing root rot incidence in chickpea followed by soil drenching and seed treatment alone.

Sindhan *et al.* (1998) found that bavistin and vitavax (0.2%) used as seed treatment, soil drenching and seed treatment plus soil drenching significantly reduced the incidence of dry root rot of chickpea with corresponding increase in phenolic compound and carbohydrate contents of the plant. Captan and thiram (0.2%) was also found quite effective in reducing disease incidence and increased the level of phenolic compounds and carbohydrate in chickpea plants. Seed treatment plus soil drenching was the most effective method of application in reducing disease incidence followed by soil drenching and seed treatment alone.

Keiwrick *et al.* (2001) reported that the combination of azox applied at 76g ai/ha and the *Bacillus* isolate MSU - 127 resulted in best disease reduction in chickpea due to *M. phaseolina*.

Resistance :

✓ Kumar and Bushan (1985) observed that liberation of total amino acids and carbohydrates in chickpea seed are correlated with the resistance and susceptibility to seed borne pathogens.

✓ Nene *et al.* (1989) found that ICC 2862, ICC 9032, ICC 10803, ICC 1155a, and ICC 11551 were resistant to root rot at 11 to 12 locations out of 18. Reddy *et al.* (1990) observed that the ICC entries 14532, 14619, 14631, 1473, 15125 and 15233 gave yield in 1988/89 in excess of 800 kg/ha. and can be used as source of resistance.

✓ Chickpea cultivar ICC 12263 was most resistant to root rot with a mean mortality rate of 2 - 3% with other entries exhibiting mean mortality rates of upto 5 per cent (Bake and Ahmed 1991).

✓ Ahmed *et al.* (1991) mortality of the susceptible entry ICC 4951 was 100% during the screening. The mean plant mortality ranged from 2.37 (in entry ICC 1263) to 5 per cent (in entry E 372).

✓ Bhatt (1993) observed that BGM 416, BG 416, ICC 1376 and IC 11314 were resistant to root rot. Resistant cultivars had greater number of total roots during early growth phases.

✓ Bejiga *et al.* (1998) observed that lines H 86-170 and ICC 89238 were resistant to *R. bataticola* at all location. Chickpea lines JG 22, JG 33, KPG 316 and Saki 93/30 showing a stable resistant reaction to root rot are suitable for use in breeding programmes (Yadava *et al.*, 2000).

MATERIAL AND METHODS

CHAPTER - III

Materials and Methods

A. Material

1. Glassware :

During the course of this investigation "Borosil" make glassware were used. They were properly cleaned with cleaning powder followed by washing with tap water and dried before use. Whenever required, the glassware were sterilized in hot air oven at 180°C for two hours. The metallic instrument such as inoculation needle used during the investigation were sterilized by heating over the flame directly.

2. Test organism :

Macrophomina phaseolina the causal organism of chickpea root rot.

3. Culture media :

I. Potato dextrose agar medium (P.D.A.) was used during the course of investigation with the following ingredients :

Peeled and sliced potato	:	200 g
Dextrose	:	20 g
Agar - agar	:	20 g
Distilled water	:	1000 ml

II. *Trichoderma* selective media (T.S.M.) :

Magnesium sulphate	:	0.2 g
K ₂ HPO ₄	:	0.9 g
Ammonium nitrate	:	1.0 g
Potassium chloride	:	0.15 g
Glucose	:	3.0 g

Agar - agar	:	15.0 g
Ridomil MZ	:	0.3 g
Chloremphenicol	:	0.2 g
Captan	:	20.0 mg
Distilled water	:	1000 ml

III) Soil maize media :

Maize (powdered)	:	200 g
Soil	:	400 g
Sand	:	400 g
Distilled water	:	1000 ml

The medium was autoclaved at a pressure of 1.05 ml/cm² for 15 minutes.

4. Seed :

The seeds of chickpea variety L 550, JG 315 and linseed variety kiran were obtained from All India Co-ordinated Research Project on chickpea, Rafi Ahmed Kidwai College of Agriculture, Sehore (M.P.).

5. Fungicide antagonists and other material :

Antagonistic organism *Trichoderma viride* was collected from National Agriculture Technology Project (RPPS 7), Department of Plant Pathology, R.A.K. College of Agriculture, Sehore (M.P.). Fungicide and other material used in present investigation were purchased from local market and from authorised dealers.

6. Meteorological data :

Data on rainfall, minimum and maximum temperature and relative humidity were obtained from the officer incharge of meteorological observatory R A K. College of Agriculture, Sehore.

Soil moisture was recorded by hardson soil moisture meter and soil temperature was recorded by inserting soil thermometer at 15 cm depth.

B. Method :

1. Procurement of chickpea dry root rot pathogen for further studies :

1.1 Survey - At first disease samples from the different places were collected from 10 villages of Sehore district. The disease intensity was calculated by putting square of 1m² made up from bamboo sticks in different randomly selected areas in the field.

1.2 Collection of diseased plant samples and symptomatology :

Dry root rot chickpea plants were collected from 10 locations in Sehore district of Madhya Pradesh at adult plant stages. Samples were brought in the laboratory and symptoms were recorded in detail. The tissue of affected plants showing the symptom of dry root rot were examined under the microscope for the presence of the causal agent.

1.3 Isolation purification and identification :

The roots of diseased chickpea plant showing root rot symptoms were washed thoroughly with tap water, small pieces from infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1 : 1000 mercuric chloride (Hgcl₂) solution for one minute followed by three changes in sterilized water to remove the trace of Hgcl₂. The pieces were then transferred aseptically to petri plates containing PDA. Inoculated petriplates were incubated

at $28 \pm 1^{\circ}\text{C}$ and examined at frequent intervals to see the growth of the fungus developing from different pieces.

1.4 Pathogenicity test :

The pathogenicity test of the fungus isolated from root rot affected chickpea plant was conducted under pot culture conditions. The fungus was multiplied in the laboratory on potato dextrose agar medium (PDA). The earthen pots (9" diameter) were filled with 3 kg soil and 15 days old inoculum of the fungus was added at the rate of 3 flasks (60 g each) in each pot. The pots were left for 7 days to allow the multiplication as well as invasion of fungus in the soil. The seeds of highly susceptible variety of chickpea (L 550) were sown at the rate of 10 seeds in each pot. The observation on germination was recorded after 10 days of sowing. While seedling mortality was recorded at fortnightly interval starting at 20 days after sowing (DAS). The dead plants were observed critically for the association of *Macrophomina phaseolina*. Isolations were made from the affected plants and the culture was compared with original one.

1.5 Growth and colony characters :

The morphological characters of pathogenic isolates of *Macrophomina phaseolina* were recorded from culture grown in PDA. Twenty ml of PDA was poured in each of the previously sterilized petriplate. Five mm discs were cut through sterilized cork borer from the margin of seven days old colony of the fungal culture growth in petri plates. One disc was placed in the centre of each plate and incubated at $28 \pm 1^{\circ}\text{C}$ for 7 days. Three replications were maintained for each isolate. The type of mycelial growth was noted as fluffy partially submerged and submerged along with the pigmentation of culture on both upper and lower surface.

1.6 Epidemiological studies :

a) Effect of weather parameters :

Observations of the weather parameters during the study period i.e. December 2003 to March 2004 were obtained from the meteorological observatory at the J.N.K.V.V., R.A.K College of Agriculture, Sehore. The weekly average of the maximum and minimum temperature and relative humidity were computed for the standard meteorological weeks.

b) Effect of edaphic parameters :

Observations of the edaphic parameters i.e. soil moisture and soil temperature were recorded at weekly intervals from the experimental site. Soil moisture was recorded by Hardson soil moisture meter and soil temperature was recorded by inserting soil thermometer at 15 cm depth in soil.

1.7 Sensitivity of different fungicide :

The variability of isolates of *Macrophomina phaseolina* was studied by recording their behaviour on six fungicides namely carbendazim (Bavistin 50% W.P.) triadimefan (Bayleton 25% W.P.), carboxin (vitavax), thiram (75% W.P.), Mancozeb (75% W.P.), and copper oxychloride 50 W.P. (Fytolan) at a concentration of 1000 ppm. Each fungicide was tested by poisoned food technique. The desired quantity of the fungicide was incorporated in the molten PDA. Three replications were maintained for each fungicide. The medium without fungicide served as control. Five mm disc cut from the margin of seven days old pure culture was transferred to each petriplate containing fungicide amended medium. The plates were then incubated at $28 \pm 1^{\circ}\text{C}$ for seven days. The diameter of the the colony was measured in mm after seven days of incubation.

2. Management of disease :

2.1 Efficacy of various seed dressing formulations against root rot of chickpea :

Design : Randomised block design

Treatments : 8

- 1. Control (untreated)
- 2. Seed treatment with carboxin + *Trichoderma viride* @ 2 g/kg + 4 g/kg
- 3. Seed treatment with carbendazim + Thiram @ 1 g/kg + 2 g/kg
- 4. Seed treatment with carboxin @ 2 g/kg
- 5. Seed treatment with carbendazim @ 1 g/kg
- 6. Seed treatment with *Trichoderma viride* @ 4 g/kg
- 7. Spraying of Dithane M 45 3 g/liter on collar region
- 8. Spraying of carbendazim @ 2 g/liter on collar region

Replication : 3

Plot size : 3 x 2.1 m

Variety : L 550

Spacing : 30 cm (Row to row)
10 cm (Plant to plant)

Date of sowing : 25/11/2003

Observation :

- a. Emergence count 15 days after sowing
- b. Incidence of root rot at regular interval of 7 days.

2.2 Evaluation of substrate of the *Trichoderma viride* :

Design : Randomised Block Design

Treatments :

1. Soil application of *Trichoderma viride* multiplied on nadep compost @ 10 kg/per plot.
2. Soil application of *Trichoderma viride* multiplied on varmi compost @ 10 kg/per plot.
3. Soil application of *Trichoderma viride* multiplied on mushroom substrate @ 10 kg/per plot.
4. Soil application of *Trichoderma viride* multiplied on field soil @ 10 kg/per plot.
5. Soil application of *Trichoderma viride* @ 4 g /plot
6. Soil treatment with Mushroom substrate @ 6.50 kg/plot
7. Soil treatment with F.Y.M. 6.50 kg/plot.
8. Control (untreated)

Replication : 3

Plot size : 3 x 2.1 m

Variety : L 550

Spacing : 30 cm (Row to row)
10 cm (Plant to plant)

Date of sowing : 25/11/2003

Observation :

- a. Emergence count 15 days after sowing.
- b. Incidence of root rot at regular interval of 7 days.

2.3 Performance of different seed treating antagonistic fungi alone and in combination of fungicide against root rot of chickpea.

Design : Randomised Block Design

Treatments : 12

1. Kanpur Str. *Trichoderma harzianum* @ 4 g/kg
2. PDBC Str. *Trichoderma harzianum* 23 @ 4 g/kg
3. PDBC Str. Thiram 10 @ 4 g/kg
4. Gulbarga Str. *Trichoderma viride* 4 g/kg
5. Carboxin @ 2 g/kg
6. T₁ + Carboxin @ 2 g/kg
7. T₂ + Carboxin @ 2 g/kg
8. T₃ + Carboxin @ 2 g/kg
9. T₄ + Carboxin @ 2 g/kg
10. Thiram + Carbendazim @ 2 + 1 g/kg
11. Thiram + Carboxin @ 2 + 2 g/kg
12. Control (Untreated)

Replication : 3

Plot size : 5 x 4 m

Variety : L 550

Date of sowing : 16/11.2003

Observation :

- a. Emergence count 15 days after sowing
- b. Incidence of root rot at regular intervals of 15, 30, 60 days.

2.4 Effect of intercropping and seed treatment on the incidence of root rot of chickpea.

Design : Factorial Randomised Block Design

Treatment : Main : 2

1. Chickpea sole crop (J.G. 315)
2. Chickpea (J.G. 315) + Linseed (Kiran) (4 : 2) intercropping

Sub treatment : 3

1. *Trichoderma viride* + *Rhizobium* @ 4 + 5 g/kg
2. Thiram + carbendazim @ 2 + 1 g/kg
3. Control (untreated)

Replication : 4

Plot size : 4.5 x 1.8 m

Variety : Chickpea (JG 315), Linseed (Kiran)

Spacing : 30 cm (Row to row)
10 cm (Plant to plant)

Date of sowing : 16/11/2003

Observation :

- a. Emergence count 15 days after sowing.
- b. Incidence of root rot at regular intervals of 15, 30, 60, 90 days.

RESULTS

CHAPTER - IV

Results

1. 1 Collection of diseased plant samples :

Total 157 samples of chickpea plants exhibiting the symptom of true root rot were collected from the cultivator fields around Sehore having the history of root rot occurrence and from the sick plot at R.A.K. College of Agriculture Sehore.

1.2 Survey on occurrence of root rot :

The data pertaining to the mean per cent incidence of root rot disease recorded during the survey at flowering stage from 10 different places around Sehore, revealed the root rot incidence ranging from 2.33 to 7.0 per cent with mean disease incidence of 4.12 per cent of the District. (Table -1). The maximum (7.0%) incidence of root rot was at Jharkhara followed by Dodi (6.66%). In other places it was below 5.0 per cent. Root rot incidence was minimum at Raipura (2.33%). Amongst the different chickpea varieties, root rot incidence was maximum 7.0 per cent in Ujjain 21, whereas, it was minimum 2.33 per cent in J G 315.

1.3 Symptomatology :

The first symptom of the disease is yellowing of the leaves within a day or two such leaves droop and in the course of the next two or three days whole plant die off. The leaves and stem of the affected plant are usually straw coloured. The tap root is dark and quite brittle in dry soil. The dark black sclerotia can be observed on and within host tissues when the stem of diseased plant is examined closely, dark lesion may be seen on the bark at the ground level. If the plants are pulled from the soil and examined, the basal stem and the main root show dry rot symptoms (Fig.1). The tissues are weakened and break off easily. In advanced cases sclerotial bodies may be seen scattered on the affected tissues.



FIG 1-SYMPTOMS OF DRY ROOT ROT OF CHICKPEA

- 1)- SPLITTED COLLAR REGION SHOWING NO INTERNAL BROWNING**
- 2)- EXTERNAL BLACKNING OF ROOT REGION EXTENDING UPTO THE COLLAR REGION (LACK OF FINER ROOTS)**
- 3)-HEALTHY ROOT AND SHOOT REGION OF CHICKPEA PLANT**

Table : 1 Root rot incidence in chickpea fields recorded at different locations of Sehore during 2003-04

Name of the village	No. of fields visited	Variety	Average root rot incidence (%)
Dodi	15	Ujjain 21	6.66
Hassanpura	13	Ujjain 24	3.66
Jamonia tank	15	Ujjain 21	5.0
Jharkhara	18	Ujjain 21	7.0
Muhali	15	Ujjain 24	3.33
Mungowali	15	J G 130	4.66
Raipura	18	Ujjain 24	2.33
Rajukhedi	15	Ujjain 24	2.66
Rola	18	J G 130	2.66
Taj	15	KAK - 2	3.33
Average	—	—	4.12

1.4 Etiology :

Isolations made from the diseased plant samples collected from different places consistently yielded a fungus exhibiting rapid, fluffy mycelial growth on potato dextrose agar medium (P.D.A.). On the basis of cultural and morphological characters the isolated purified fungus was identified as *Macrophomina phaseolina* (Tassi) Goid on the basis of the standard mycological keys and pathogenicity on chickpea. The fungus invaded the host both inter and intracellularly. It grew rather fast, covering large areas of the host tissue and eventually killing them in a short time. It produces numerous sclerotial bodies on the host tissue measuring 110-130 μ in diameter (Fig.2a). Often the conidial or pycnidial stage (*Macrophomina phaseolina*) is produced on the host. The pycnidia are dark brown, ostiolate and of varying size depending upon the host. The pycniospores are ellipical thin walled, single celled, hyaline and measure 10 - 42 x 6 - 104. (Fig 2b)

1.5 Pathogenicity :

The pathogenicity of the *Macrophomina phaseolina* (Tassi) Goid, *Rhizoctonia bataticola* (Taub) Butler isolated from the affected chickpea plants confirmed by multiplying the fungus and mixing it in sterile sand in pots. Symptoms of root rot appeared at 15-20 days in susceptible variety L-550. The affected plants exhibited sudden drying and root rot symptom. The total final mortality was 95 per cent. Re isolation from the roots of the root rot affected plants consistently yielded *Rhizoctonia bataticola* Sclerotial stage of the fungus *M. phaseolina*, thus confirming the pathogenic potential of the isolated fungus.

2. Epidemiological studies :

Observations of the weather parameters of the study period i.e. December 2003 to March, 2004 were obtained from the meteorological observatory of the J.N.K.V.V., R.A.K. College of Agriculture, Sehore. Observations of the edaphic

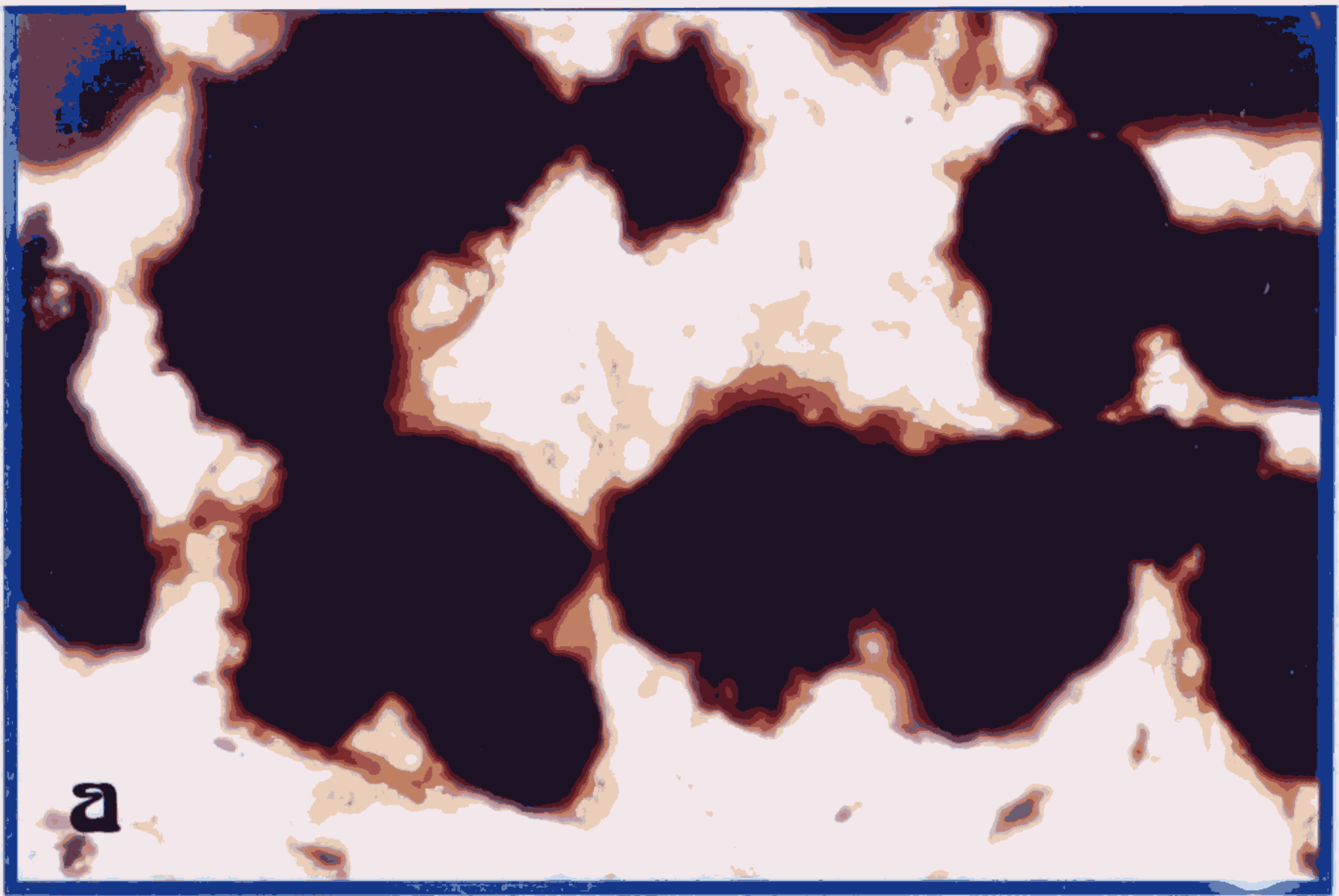


FIG2-(a) Sclerotial stage
(Rhizoctonia bataticola)
 showing irregular shape
 sclerotia

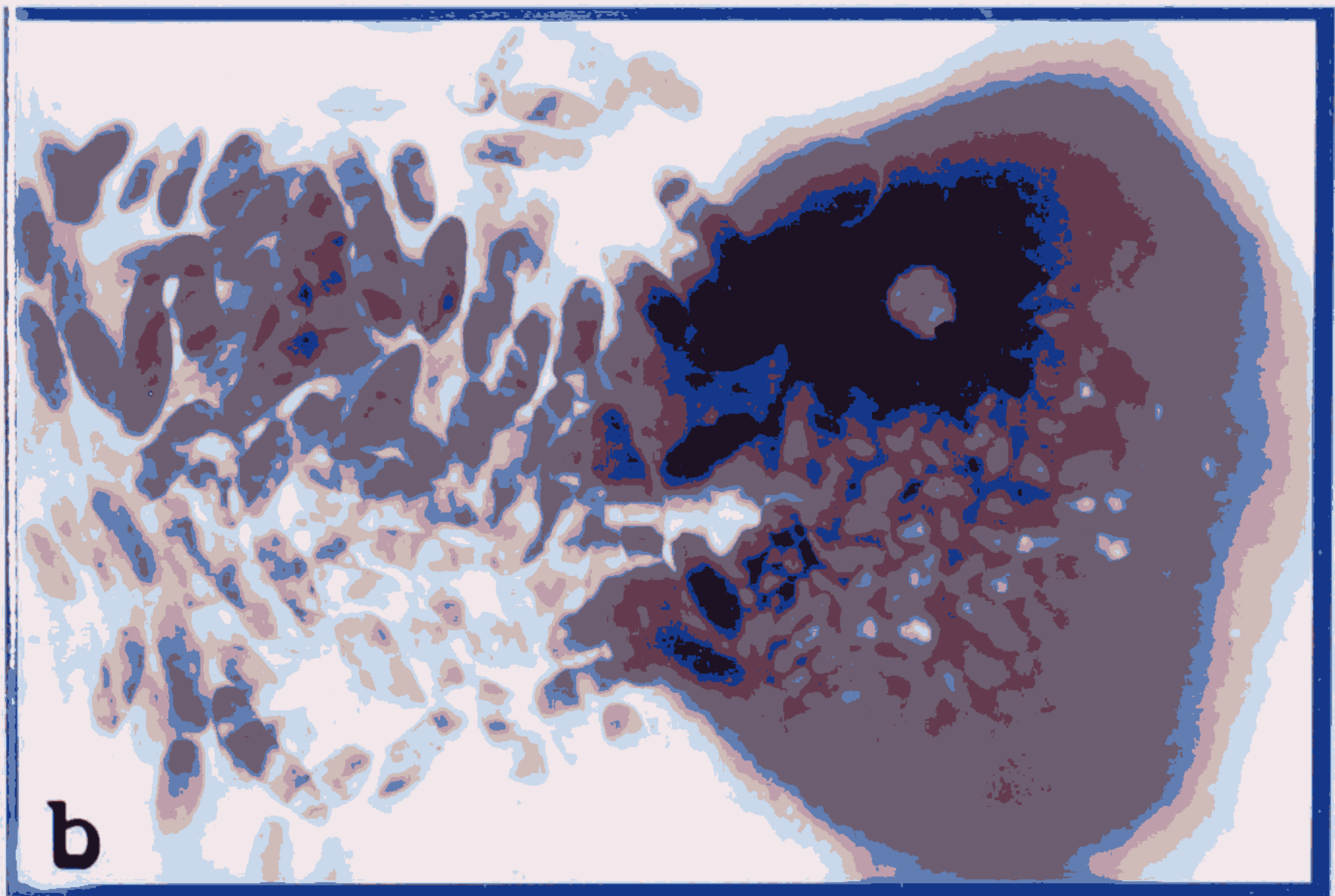


FIG2(b) Pycnidial stage
(Macrophomina phaseolina)
 showing pycnidium and
 pycnidiospores

parameters ie. soil moisture and soil temperature were recorded at weekly intervals from the experiment site as described earlier. The weekly average of the maximum and minimum temperature and relative humidity were computed for the standard meteorological weeks (Table 2 and 3).

2.1 Effect of meteorological parameters on the chickpea root rot incidence:

The development of root rot incidence was studied in relation to various climatic parameters. The absolute root rot incidence recorded every week was correlated with the mean maximum and minimum temperature and relative humidity of the corresponding meteorological weeks and correlation coefficient (r) value were computed for each parameter and given in Table 4. To quantify the effect of various parameters the linear regression equation was also worked out. The value of correlation coefficient (r) arrived for various combination indicated that the major climatic parameters maximum and minimum temperature had a significant relationship with root rot incidence while relative humidity exhibited a negative relationship with the root rot incidence in chickpea.

2.1.1. Maximum temperature :

Maximum temperature was found to have a positive correlation with chickpea root rot incidence relationship being significant ($r = 0.66$). The regression equation showed that root rot incidence increased by 0.63% with increase of 1°C of maximum temperature. (Fig.3).

2.1.2. Minimum temperature :

Minimum temperature was found to have a positive correlation with chickpea root rot incidence relationship being significant ($r = 0.65$). The

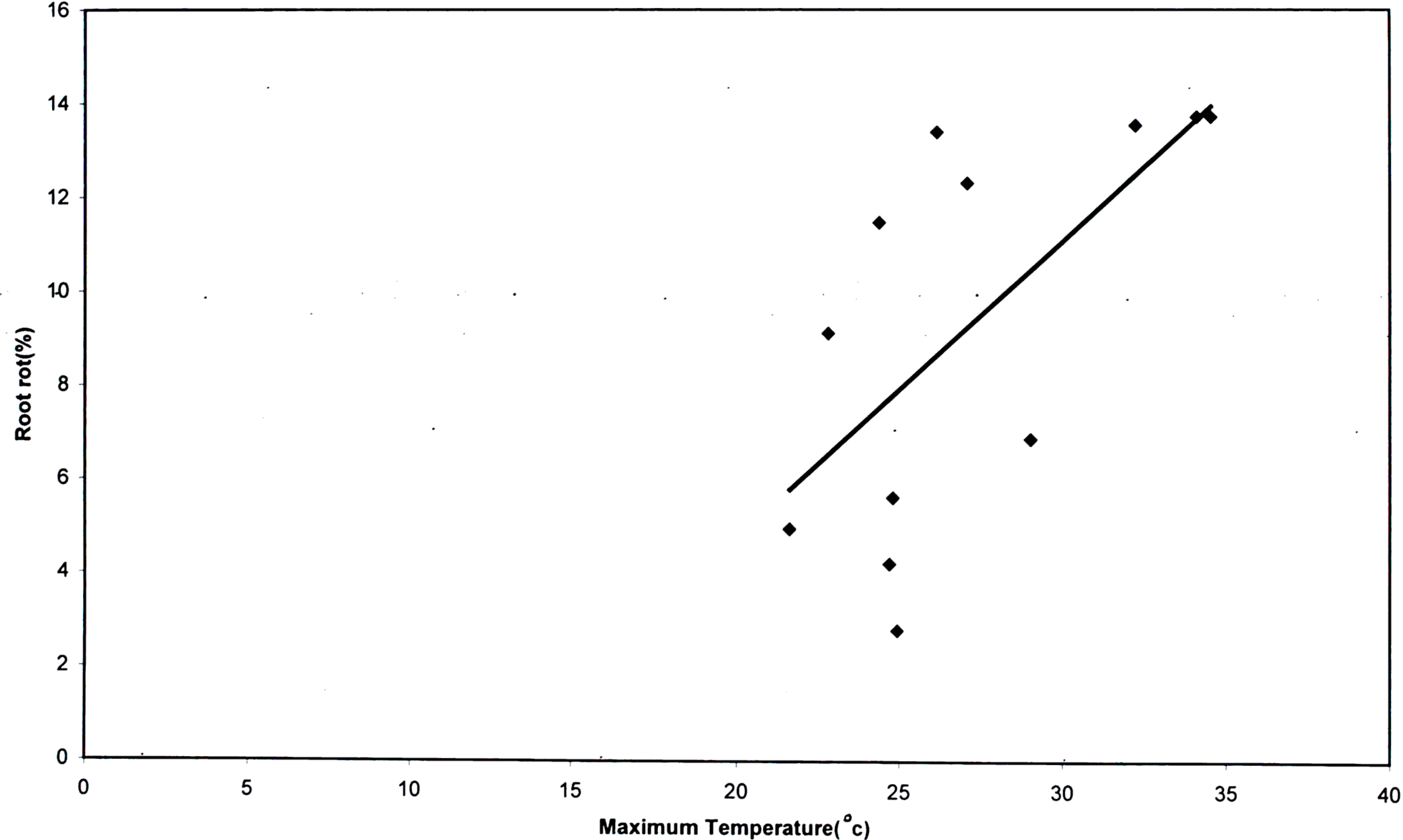
Table 2 : Weakly observations on meteorological parameters and root rot incidence (2003-04)

Period	Mean weekly temperature		Relative humidity (%)	Root rot (%)
	Minimum °C	Maximum °C		
17-23 Dec. 2003	5.86	24.93	76.29	2.79
24-31 Dec. 2003	4.66	24.69	75.25	4.21
01-07 Jan 2004	2.93	21.60	78.57	4.95
08-14 Jan 2004	9.86	24.79	69.57	5.62
15-21 Jan 2004	12.07	29.00	63.57	6.87
22-28 Jan 2004	10.57	22.79	69.14	9.12
29 Jan - Feb. 04	9.00	24.36	72.00	11.48
05 - 11 Feb. 04	7.64	27.07	68.43	12.32
12-18 Feb. 04	9.14	26.14	68.57	13.40
19-25 Feb. 04	12.21	32.21	65.29	13.55
26 Feb- March, 04	13.79	34.07	58.43	13.73
4-10 March, 04	12.07	34.50	53.57	13.73

Table 3 : Weekly observations on soil edaphic parameters and root rot incidence in chickpea (2003-04)

Period	Soil moisture (%)	Soil temperature (%)	Root rot (%)
17-23 Dec. 2003	60.82	19.28	2.79
24-31 Dec. 2003	73.42	16.10	4.21
01-07 Jan 2004	89.85	17.00	4.95
08-14 Jan 2004	88.71	18.70	5.62
15-21 Jan 2004	86.28	19.41	6.87
22-28 Jan 2004	83.62	19.28	9.12
29 Jan - 04 Feb	82.00	19.00	11.48
05 - 11 Feb	79.85	21.57	12.32
12-18 Feb	77.71	23.71	13.40
19-25 Feb	75.00	23.85	13.55
26 Feb-03 March	72.00	25	13.73
4-10 March	69.00	25	13.73

Fig 3 : Relationship of root rot of chickpea with Maximum Temperature



regression equations indicated that with every 1°C increase in minimum temperature the disease incidence increased by 0.83% (Fig. 4).

2.1.3. Relative humidity :

Relative humidity was found to have a negative correlation with chickpea root rot incidence and had significant correlation ($r = -0.72$). The regression equation indicated that with the every 1 per cent increase in relative humidity the disease incidence decreased by 0.41 per cent (Fig. 5).

2.2 Effect of edaphic parameters on the chickpea root rot incidence :

The development of root rot incidence was studied in relation to various edaphic parameters. The absolute disease incidence recorded every week was correlated with the mean soil moisture and soil temperature of the corresponding meteorological weeks and correlation coefficient (r) value were computed for each parameters and are given in (Table 4). To quantify the effect of various combination indicated that the major edaphic parameter soil temperature had a significant relationship with root rot incidence.

2.2.1 Soil temperature :

Soil temperature was found to have a positive correlation with chickpea root rot incidence relationship being significant ($r = 0.83$). The regression equation indicated that with the every 1°C increase in soil temperature the disease incidence increased by 1.14 per cent (Fig. 6).

2.2.2. Soil moisture :

Soil moisture was found to have a negative correlation with chickpea root rot incidence and it was non significant ($r = -0.11$). The regression equation

Table 4 : Correlation coefficient (r) values between incidence of chickpea root rot and various weather parameters

Weather/Edaphic parameters	Correlation coefficient
Maximum temperature	0.66 *
Minimum temperature	0.65 *
Relative humidity	-0.72 *
Soil temperature	0.83 *
Soil moisture	-0.11

* Significant at 5.0 per cent.

Table 5 : Regression equations between incidence of chickpea root rot and various weather parameters

Weather parameters	Regression equations
Maximum temperature	$Y = 0.63 x - 7.86$
Minimum temperature	$Y = 0.83 x - 1.67$
Relative humidity	$Y = -0.41 x - 37.28$
Soil temperature	$Y = 1.14 x - 14.43$
Soil moisture	$Y = -0.052 x + 13.43$

Fig. 4 : Relationship of root rot of chickpea with Minimum Temperature

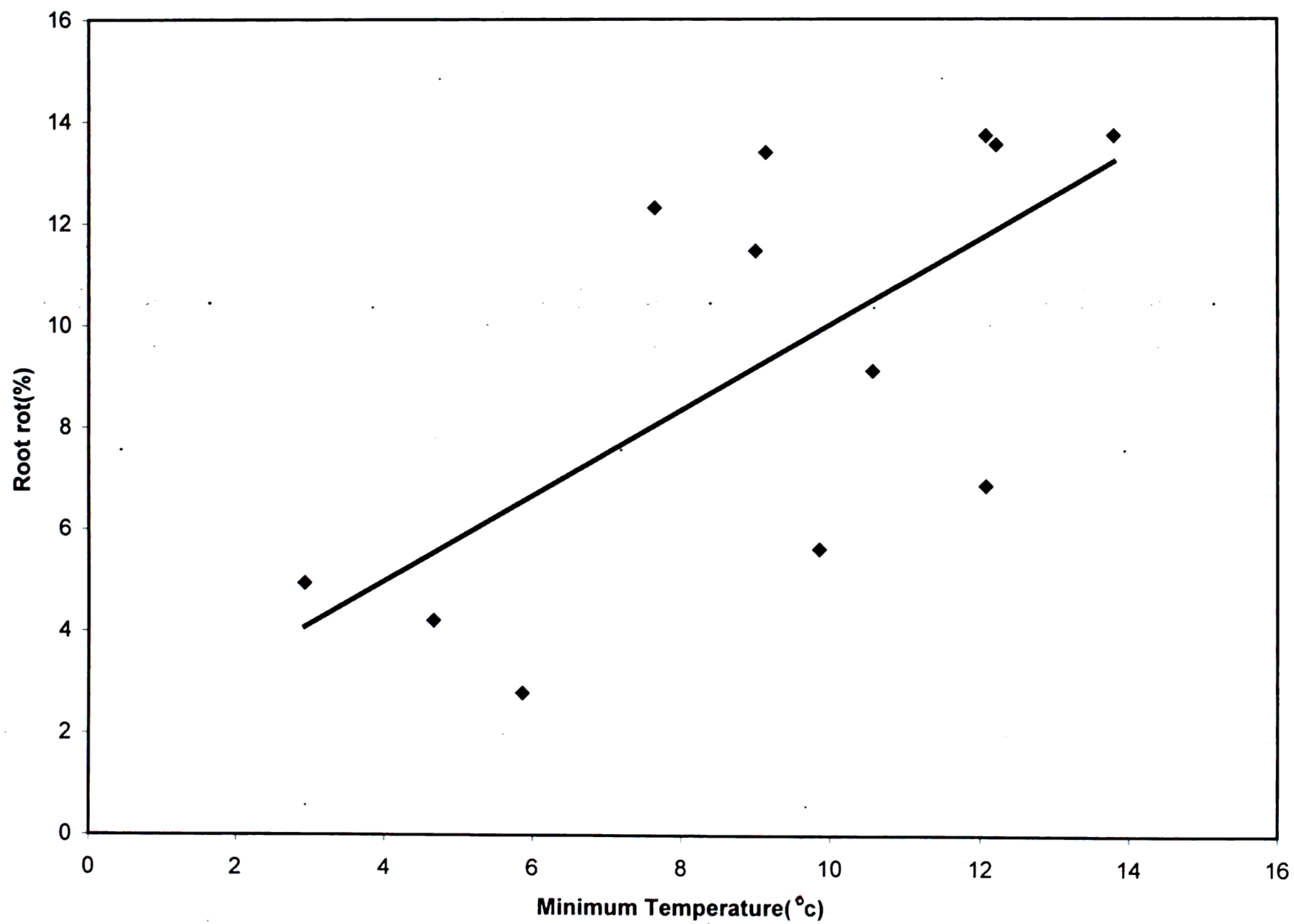


Fig 5 : Relationship of root rot of chickpea with Relative Humidity

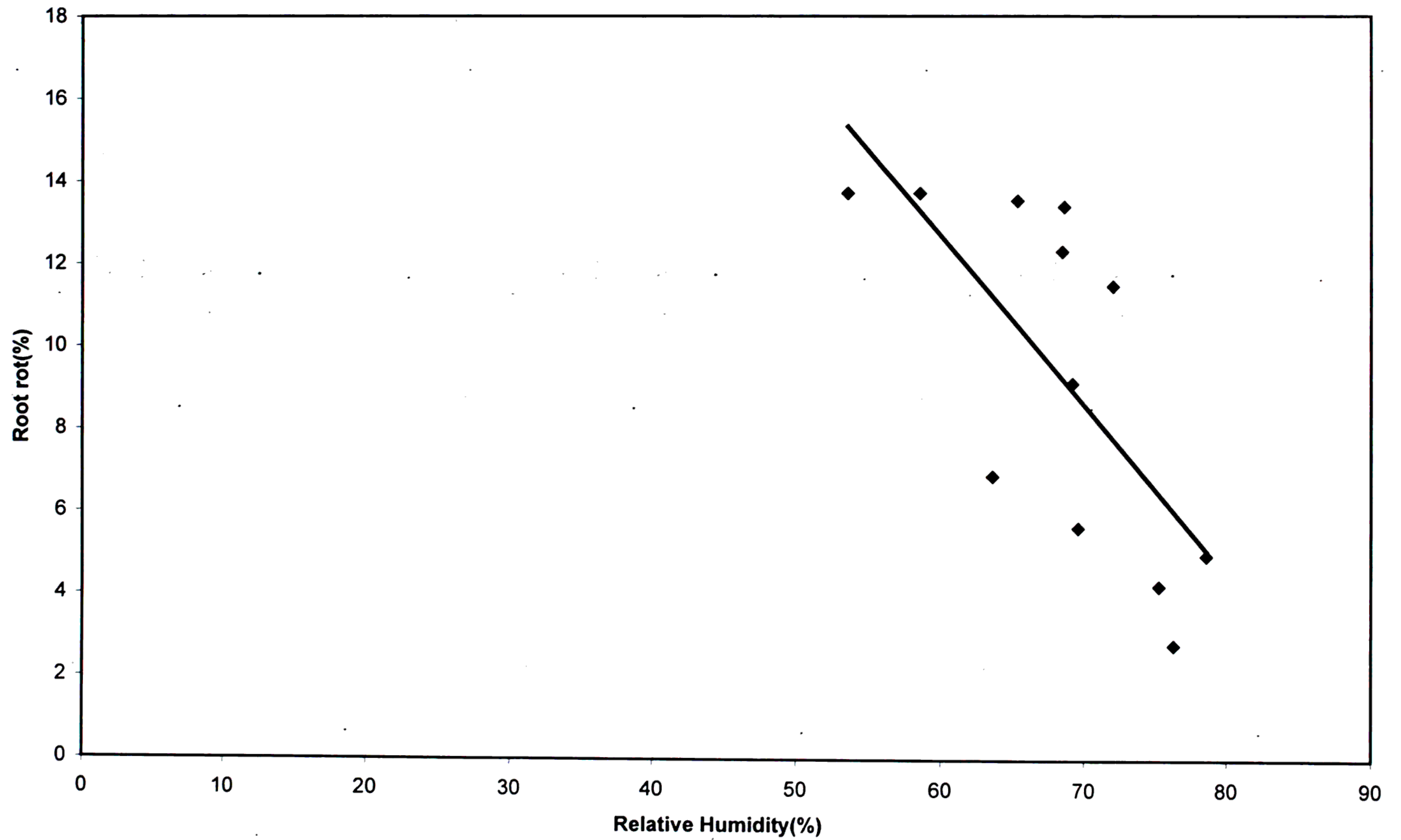
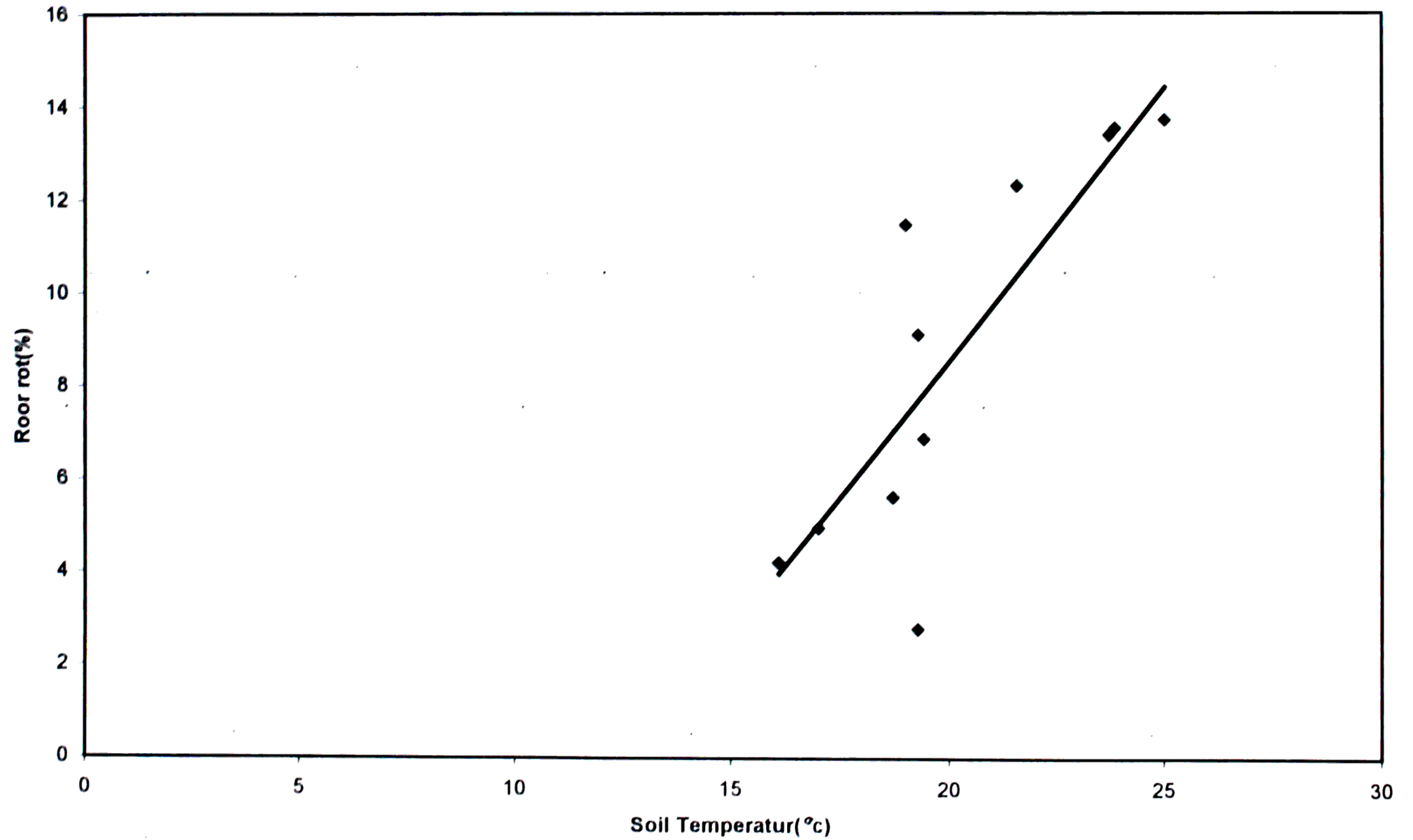


Fig 6 : Relationship of root rot of chickpea with Soil Temperature



indicated that with the every 1 per cent increase in soil moisture the disease incidence decreased by 0.05 per cent.

3. Management :

3.1 Effect of fungicides and bioagents on root rot incidence and yield :

The incidence of chickpea root rot was affected by various seed treating agents and all seed treatments were able to reduce the root rot incidence to significant levels. Seed treatment with thiram + carbendazim @ 2 + 1g/kg and carboxin 2 g/kg were most effective and with lowest (1.8%) root rot incidence and was followed by seed treatment with thiram + carboxin (2.2%) Project Directorate of Biological Control (PDBC) Str. *Trichoderma harzianum* (2.5%) Kanpur Str. *Trichoderma harzianum* + Carboxin (2.6%), PDBC Str. *T. harzianum* + Carboxin (2.7%), PDBC Str Thiram (3.7%), PDBC Str. Thiram + carboxin (3.8%), *T. viride* + carboxin (4.2%), *T. viride* (5%) and Kanpur Str. *T. harzianum* (5.1%). (Table 6).

Chickpea yield was significantly increased by seed treatment Thiram + carbendazim, Thiram + carboxin, carboxin, PDBC Str. *T. harzianum* + carboxin and PDBC Str. Thiram + carboxin. The chickpea yields were not significantly influenced by seed treatment with Gulbarga Str. *T. viride*, PDBC Str. Thiram, Kanpur Str. *harzianum*, Gulbarga Str. *T. viride* + carboxin, Kanpur Str. *T. harzianum* + carboxin PDBC Str. *T. harzianum*. Highest yield of 1527 kg/ha was recorded when the seeds were treated with Thiram + carbendazim (Table 6). Highest net return of Rs. 3776.5/ha was obtained under conventional seed treatment with Thiram + carbendazim (2 : 1), followed by treatment with Thiram + Carboxin (1 : 1) (Rs. 3007.0) carboxin (Rs. 2684.5/ha), PDBC Str. *T. harzianum* + carboxin (2 : 1) (Rs. 2488.5/ha), PDBC Str. Thiram + carboxin (2 : 1) (Rs. 2416.0), Kanpur Str. *T. harzianum* + carboxin (2 : 1) (Rs. 2198.5), Gulberga Str. *T. viride* + carboxin (2 : 1) (Rs. 2082.5), Kanpur Str. *T. harzianum* (Rs. 1892.0), PDBC Str. Thiram (Rs. 1848.5), Gulberga Str. *T. viride* (Rs. 746.5) (Table 7).

Table : 6 Root rot incidence (%) and yield of chickpea (kg/ha) in different seed treating agents

Treatments	Root rot incidence (%)	Yield (kg/ha)
Kanpur Str. <i>Trichoderma harzianum</i> (T ₁)	5.1 (13.04)	1396
PDBC Str. <i>Trichoderma harzianum</i> (T ₂)	2.5 (9.15)	1433
PDBC Str. Thiram (T ₃)	3.7 (10.88)	1393
Gulbarga Str. <i>Trichoderma viride</i> (T ₄)	5.0 (12.91)	1317
Carboxin	1.8 (7.68)	1461
T ₁ + Carboxin	2.6 (9.15)	1461
T ₂ + Carboxin	2.7 (9.44)	1453
T ₃ + Carboxin	3.8 (11.03)	1448
T ₄ + Carboxin	4.2 (11.77)	1425
Thiram + carbendazim	1.8 (7.68)	1527
Thiram + carboxin	2.2 (8.57)	1486
Control (Untreated)	8.2 (16.64)	1260
SEm ±	0.55	23.52
CD at 5%	1.61	68.97

Figures in parantheses are Arcsin transfored values.

Fig. 7 : Root rot incidence (%) and yield of chickpea (q/ha) in different seed treating agents

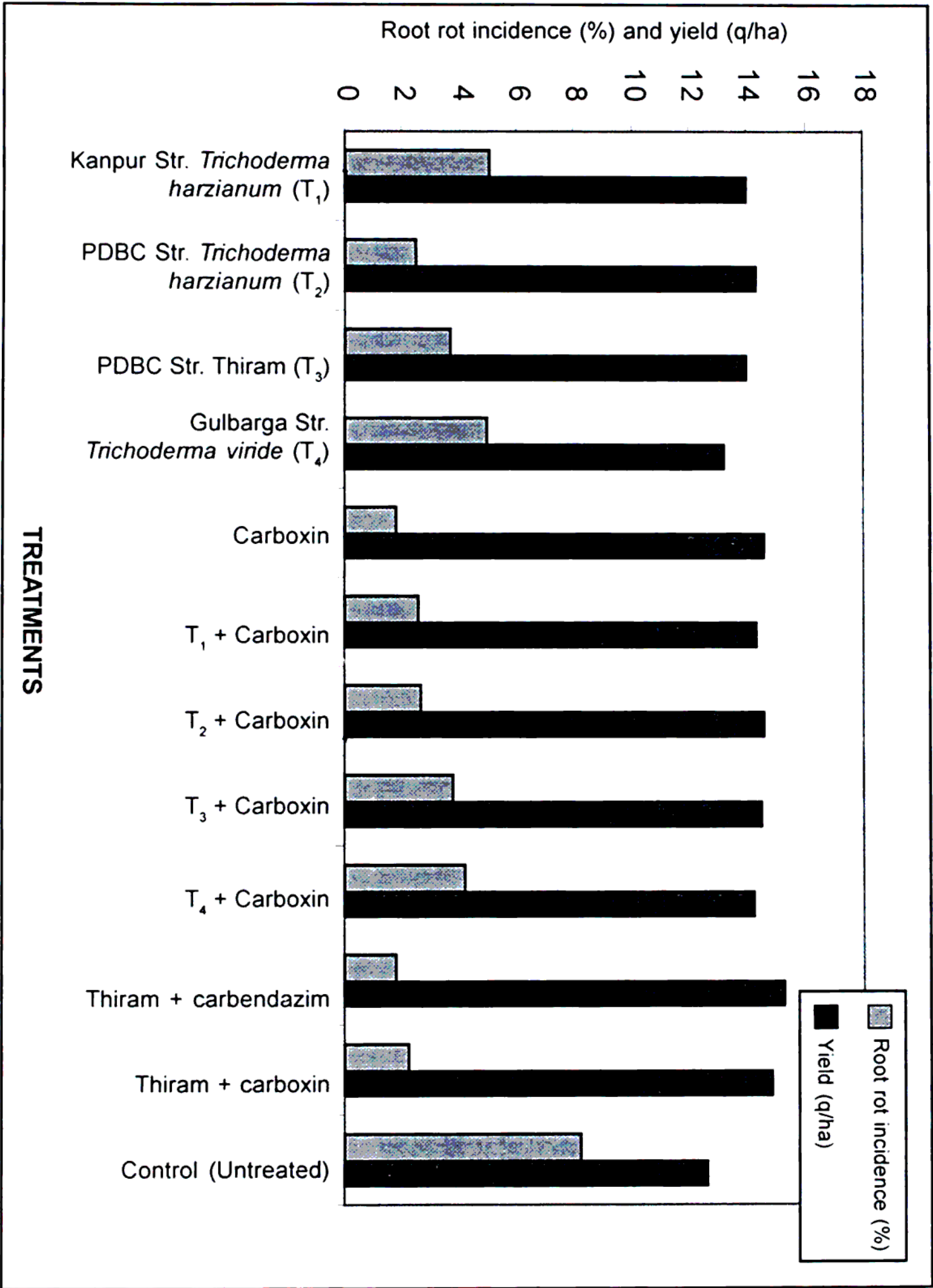


Table : 7 Effect of various indigenously used seed treating agents on the chickpea yield, net return and cost benefit ratio

Treatments	Dose g/ha	Yield kg/ha	Increase in yield over control kg/ha	Value of additional yield Rs./ha	Treatment cost Rs/ha	Net return Rs/ha	C : B ratio
Kanpur Str. <i>Trichoderma Harzianum</i> (T ₁)	400	1396	136	1972.0	80	1892.0	1 : 23.65
PDBC Str. <i>Trichoderma Harzianum</i> (T ₂)	400	1633	173	2508.5	80	2428.5	1 : 30.35
PDBC Str. Thiram (T ₃)	400	1393	133	1928.5	80	1848.5	1 : 23.10
Gulbarga Str. <i>Trichoderma viride</i> (T ₄)	400	1317	57	826.5	80	746.5	1 : 9.70
Carboxin	200	1461	201	2914.5	230	2684.5	1 : 11.67
T ₁ + Carboxin	400 + 200	1433	173	2508.5	310	2198.5	1 : 7.09
T ₂ + Carboxin	400 + 200	1453	193	2798.5	310	2428.5	1 : 8.02
T ₃ + Carboxin	400 + 200	1448	188	2726.0	310	2416.0	1 : 7.79
T ₄ + Carboxin	400 + 200	1425	165	2392.5	310	2082.5	1 : 6.71
Thiram + carbendazim	200 + 100	1527	267	3871.5	95	3776.5	1 : 39.75
Thiram + carboxin	200 + 200	1436	226	3277.0	270	3007.0	1 : 11.13
Control (Untreated)	--	1260	--	--	--	--	--
SEm ±		23.52					
CD at 5%		68.97					

Chickpea market price 1450/q. Treatment cost includes the cost of various seed treating agents @ Rs. 20/100 g *Trichoderma harzianum* Rs. 20/100 g Thiram, Rs. 115/100 g carboxin (as vitavax), Rs. 55/100 g carbendazim (as Bavastin) Rs. 20/100 g *Trichoderma viride*.

The cost benefit ratio under seed treatment with Thiram + carbendazim was quite high 1 : 39.75 with highest yield and net return. Seed treatment with PDBC Str. *T. harzianum* with C : B ratio of 1 : 30.35 was next in order (Table 7).

3.2 Effect of *Trichoderma viride* multiplied on different substrates on root rot incidence :

The incidence of chickpea root rot was also affected by soil application of *T. viride* multiplied on different substrates but only three of them were effective in reducing the root rot incidence to significant levels. Soil application with *T. viride* multiplied on vermi compost was most effective with lowest (3.6%) root rot incidence and followed by treatment *T. viride* multiplied on mushroom substrate (4.2%) and seed treatment with *T. viride* (5.4%) was next to them (Table -8). As obvious the chickpea yield was significantly increased in *T. viride* multiplied on varmi compost. *Trichoderma viride* multiplied on mushroom substrate. Seed treatment *T. viride* and *T. viride* multiplied on field soil were equally effective in reducing root rot of chickpea as compared to control.

Highest yield 994 kg/ha was recorded under treatment with *T. viride* multiplied on varmi compost. (Table -8).

Highest net return of Rs. 7461.5/ha was obtained in soil application *T. viride* multiplied on vermi compost followed by seed treatment with *T. viride* multiplied on mushroom substrate (Rs. 6861.5), seed treatment *T. viride* (Rs. 6483.0), (C : B) Cost Benefit Ratio in seed treatment with *T. viride* was maximum 1 : 64.83 with high yield and net return. Application of *T. viride* multiplied on mushroom substrate gave C : B ratio of 1 : 34.30 was next in order (Table -9).

3.3 Effect of application of fungicides and bio-agent on root rot incidence and yield of chickpea :

The incidence of chickpea root rot was also affected by various treatment and all the treatments were able to reduce the root rot incidence to significant

Table 8 : Root rot incidence (%) and yield of chickpea (kg/ha) in different substrates

Treatments	Root rot incidence (%)	Yield (kg/ha)
<i>Trichoderma viride</i> on Nadep compost	6.7 (14.9)	553
<i>Trichoderma viride</i> on vermi compost	3.6 (10.8)	994
<i>Trichoderma viride</i> on mushroom substrate	4.2 (11.8)	894
<i>Trichoderma viride</i> on field soil	6.1 (14.1)	741
Seed treatment by <i>Trichoderma viride</i>	5.4 (13.3)	861
Mushroom substrate	6.1 (14.2)	680
<i>Trichoderma viride</i> on farm yard manure	6.2 (14.3)	658
Control	8.4 (16.8)	407
SEm ±	1.23	91.40
CD at 5%	2.64	277.25

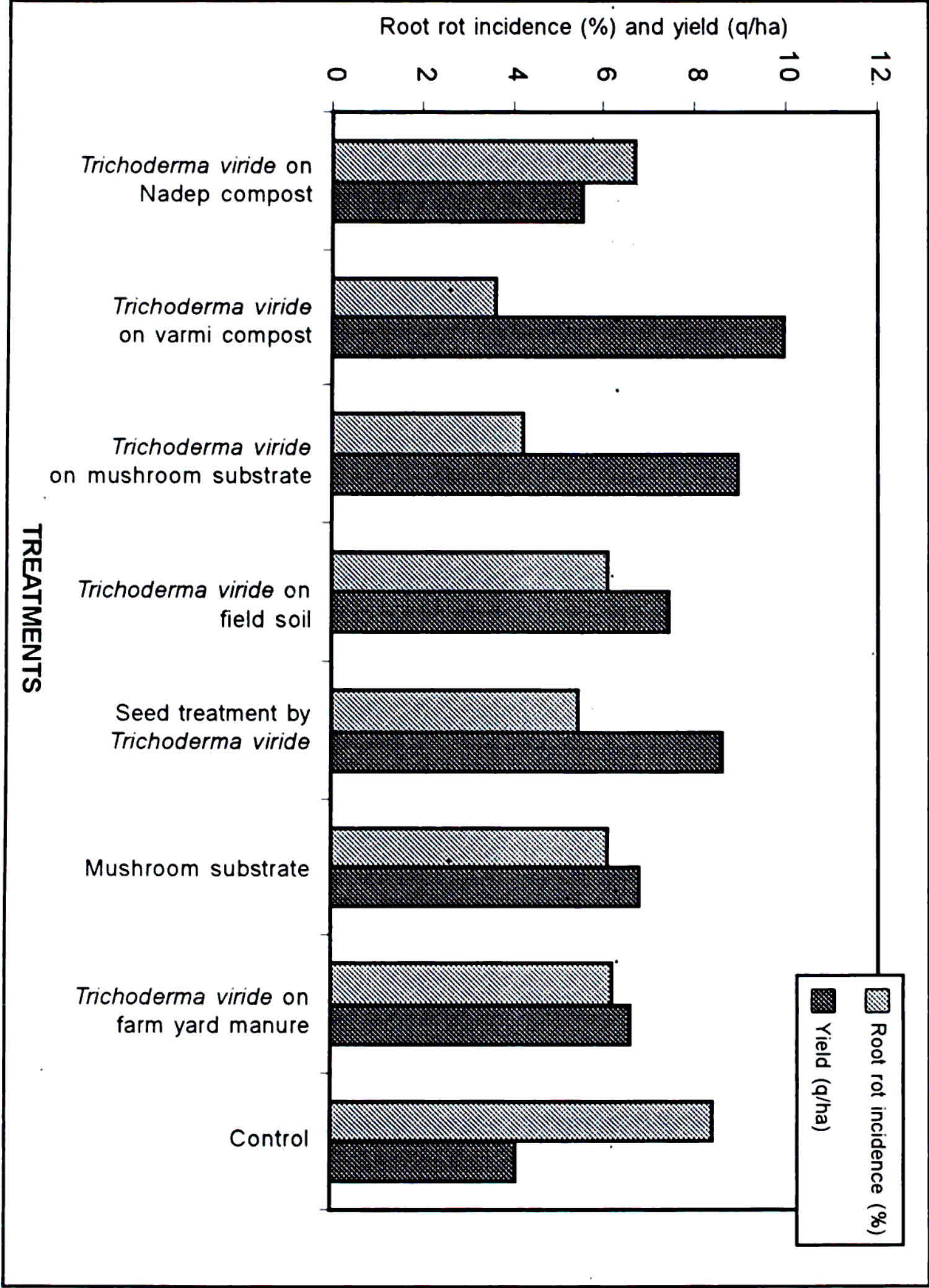
Data in parantheses are Arcsin $\sqrt{\text{percentage}}$ transformed values.

Table : 9 Effect of *Trichoderma viride* multiplied on different substrates on the chickpea yield, net return and cost benefit ratio.

Treatments	Dose g/ha	Yield kg/ha	Value of additional yield Rs./ha	Treatment cost Rs/ha	Net return Rs/ha	C : B ratio
<i>Trichoderma viride</i> on Nadep compost	250 q + 2.5 kg	553	2117.0	1750.0	367.0	1 : 0.20
<i>Trichoderma viride</i> on vermi compost	2.5 q + 250 g	994	8511.0	1050.0	7461.5	1 : 7.10
<i>Trichoderma viride</i> on mushroom substrate	2.5 q + 250 g	894	7061.5	200.0	6861.5	1 : 34.30
<i>Trichoderma viride</i> on field soil	20 kg + 20 kg	741	4843.0	4000.0	843.0	1 : 0.21
Seed treatment by <i>Trichoderma viride</i>	500 gm	861	6583.0	100.0	6483.0	1 : 64.83
Mushroom substrate	2.5 q	680	3958.5	150.0	3808.5	1 : 25.39
<i>Trichoderma viride</i> on farm yard manure	25 q + 2.5 kg	658	3639.5	1750.0	1889.5	1 : 1.07
Control	—	407	—	—	—	—
SEm ±		91.40				
CD at 5%		277.25				

Chickpea market price 1450/q. Treatment cost include the cost of various seed treating agents @ Rs. 50/q (Nadep compost), Rs. 50/q (Farm yard manure), Rs. 400/q (Vermi compost), Rs. 60/q (Mushroom substrate), Rs. 200/kg (*Trichoderma viride*).

Fig. 8 : Effect of different substrates of *Trichoderma viride* on root rot incidence and yield of chickpea.



levels (Fig. 8). Seed treatment with carbendazim + Thiram and *Trichoderma viride* were most effective with lowest (10.2%) root rot incidence and followed by seed treatment with carboxin + *T. viride* (10.7%). Chickpea yield was significantly increased by seed treatment carbendazim + thiram, *T. viride*, carboxin and spray carbendazim on collar region of the plants. Highest yield of 1011.4 kg/ha was recorded under seed treatment with carbendazim + Thiram (Table -10).

Highest net return (Table 11) of Rs. 7648.0/ha was obtained under conventional seed treatment with carbendazim + thiram followed by seed treatment with *T. viride* (Rs. 6894.5), carboxin + *T. viride* (Rs. 6215.0).

Cost benefit (C : B) ratio under seed treatment with carbendazim + thiram was quite high 1 : 80.50 with highest yield and net return. Seed treatment with *T. viride* with C : B ratio of 1 : 80.18 was next in order (Table 11).

4.1 Evaluation of integrated disease management (I.D.M.) component against chickpea root rot :

Root rot incidence :

There was no significant effect of linseed intercropping on root rot incidence in chickpea.

Seed treatment with fungicide (Thiram + carbendazim) bio control agent (*Trichoderma viride* + *Rhizobium*) both significantly reduced the chickpea root rot incidence (Table 12).

Cobination of both thiram + carbendazim was effective in reducing the root rot incidence to 7.69% in comparison to 12.15% recorded under untreated control. Seed treatment with *T. viride* + *Rhizobium* @ 4 + 5 g/kg with 9.02% disease incidence was least effective.

Table 10 : Root rot incidence (%) and yield of chickpea (kg/ha) in different applications of fungicides and bio agents

Treatments	Root rot incidence (%)	Yield (kg/ha)
Corboxin + <i>Trichoderma viride</i>	10.70 (19.00)	926.8
Carbendazim + Thiram	10.20 (18.63)	1011.4
Carboxin	10.80 (19.19)	791.6
Carbendazim	13.10 (21.22)	737.4
<i>Trichoderma viride</i>	10.20 (20.18)	959.5
Spray of carbendazim on collar region of 1.5 month crop	11.90 (20.18)	771.9
Spray of Dithane M - 45	13.60 (21.64)	698.9
Control	26.10 (30.72)	477.1
SEm ±	0.69	81.47
CD at 5%	2.08	247.14

Data in paranthesis are sin $\sqrt{\text{percentage}}$ transformed values.

Table : 11 Effect of different applications of fungicides and bio agents on chickpea yield, net return and cost benefit ratio.

Treatments	Dose g/ha	Yield kg/ha	Increase in yield over control kg/ha	Value of additional yield Rs./ha	Treatment cost Rs/ha	Net return Rs/ha	C : B ratio
Carboxin + <i>Trichoderma viride</i>	200 + 400	926.8	450	6525.0	310	6215.0	1 : 20.04
Carbendazim + Thiram	100 + 200	1011.4	534	7743.0	95	7648.0	1 : 80.50
Carboxin	200	791.6	315	4567.5	230	4337.5	1 : 18.85
Carbendazim	100	737.4	260	3770.0	55	3715.0	1 : 67.54
<i>Trichoderma viride</i>	400	959.7	481	6974.5	80	6894.5	1 : 80.1
Spray of carbandazim on collar region of 1.5 month crop	2000	771.9	295	4277.5	1100	4222.5	1 : 3.83
Spray of Dithane M - 45	2000	698.9	222	3219.0	1250	3094.0	1 : 2.47
Control	—	477.1	—	—	—	—	—
SEm ±		81.47					
CD at 5%		247.14					

Chickpea market price 1450/q. Treatment cost include the cost of various seed treating agents @ 20/100 g (Thiram), Rs. 20/100 g (*Trichoderma viride*), Rs. 115/100g Carboxin (as vitavax), Rs. 55/100 g carbendazim (as Bavistin)

Fig. 9 : Effect of different seed treatments and spraying on root rot incidence and yield of chickpea

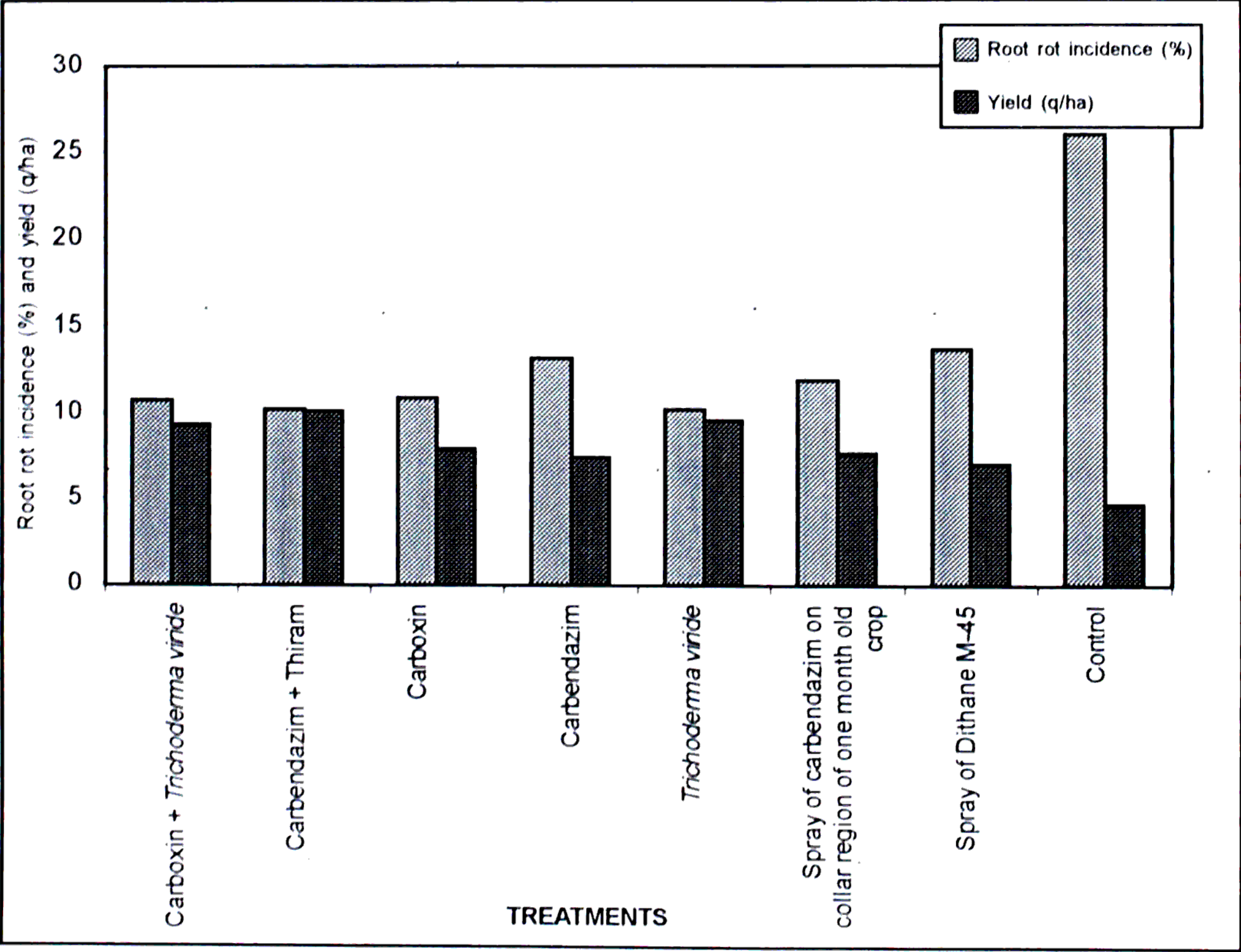


Table : 12 Effect of different components in integrated disease management (IDM) on the incidence of root rot of chickpea and its yield.

Treatments	Root rot incidence (%)	Yield (kg/ha)
Main treatment		
1) Chickpea Sole	9.54	1248.96
2) Chickpea + linseed (4 : 2)	9.71	1600.61
SEm ±	0.22	16.39
CD at 5%	NS	NS
Sub. Treatment		
1) <i>Trichoderma viride</i> + <i>Rhizobium</i> @ 4 + 5 g/kg	9.02	1431.35
2) Thiram + carbendazim @ 2 + 1 g/kg	7.69	1565.79
3) Control (untreated)	12.15	1277.36
SEm ±	0.27	20.07
CD at 5%	0.84	61.84
Interation M x S	NS	NS

The interaction of cropping system with seed treatment was non significant for root rot incidence. The chickpea yield was not significantly influenced by various intercropping system though chickpea + linseed (4 : 2) intercropping had higher chickpea equivalent yield of 1600.61 kg/ha in comparison to 1248.96 kg/ha recorded under chickpea sole crop. The chickpea equivalent yield was significantly increased by all the seed treatment. Highest yield of 1565.79 kg/ha was recorded under combination of both the seed treatment in thiram (2 g/kg) + carbandazim (1 g/kg), followed by *T. viride* (4 g/kg) + Rhizobium (5 g/kg). The interaction of cropping system x seed treatment was also not significant for chickpea yield (Table 12).

5.1 Resistance in chickpea lines to *Macrophomina phaseolina* (Tassi) Goid in Sick plot :

Screening of 80 lines was done in wilt/root rot sick plot of chickpea the data are presented in table 13. The percentage of root rot incidence was recorded chickpea (47) showing different reactions were as following :

Fortyseven varieties viz., PG 9426-2, KH 94-134, HKOO 299, IPC 305, NPJGK 11, GJG 2018, GJG 9905, IPC 98-22, PG 9425-5, BG 372, BG 1103, BGM 547, GNG 1488, H 82-2, PBG 1, PG 9425-9, IPC 98-12, CST 104, RSG 143-1, RSG 7-931, PG 9425-5, BG 1053, BG 1108, BG 2024, MPJGK 6, PG 97313, PG 95333, CST 103, GNG 508, IPC 99-12, GNG 1519, HHOO 191, IPC 2001-01, IPC 2001-12, MPJG 99-213, H 82-2, HOO 65, IPC 97-29, IPC 2000-27, GNG 1503, GNG 1515, HOO 237, HH 99-265, IPC 94-132, BGD 128, MPJGK 99-115, Phule G 96006 showed root rot incidence less than 10 per cent.

Six varieties IPC 2k-25, IPC 97-67, L 550, PBKG 220, CST 140, CST 160 showed root rot incidence between 10 to 20 per cent.

Three varieties RSG 963, RSG 888, GNG 1499 showed root rot more than 20 per cent.

In susceptible check L 550 root rot incidence was 27.62 per cent.

Table 13 : Performance of chickpea entries against root rot during 2003-2004

S.No.	Entries	Root rot (%) Mean
1.	CST 103	9.70
2.	GNG 508	0.70
3.	GNGN 1512	0.00
4.	HOO 78	0.00
5.	HOO 126	0.00
6.	IPC 99-12	0.73
7.	MPJG 2001-02	0.00
8.	WCG 2000 -01	0.00
9.	IPC 2000 - 52	0.00
10.	GNG 1477	0.00
11.	GNG 1519	0.95
12.	HOO 212	0.00
13.	HHOO 191	4.75
14.	IPC 2001-01	0.90
15.	IPC 2001-12	1.75
16.	MPJG 99-213	1.35
17.	PG 9421-1	0.00
18.	WCG 2000-02	0.00
19.	GNG 1533	0.00
20.	H 82-2	0.65
21.	HOO 65	1.50
22.	HH 99-291	0.00
23.	IPC 97-29	0.55
24.	IPC 2000-27	3.05
25.	PG 97030	0.00
26.	GNG 1503	0.65
27.	GNG 1515	1.00
28.	HOO 237	0.95

S.No.	Entries	Root rot (%) Mean
29.	HH 99-265	1.40
30.	IPC 94 - 132	0.70
31.	IPC 2001-02	0.00
32.	PG 9426-2	0.75
33.	VIJAY	0.00
34.	HK 94-134	3.48
35.	HK 98-155	0.00
36.	HKOO 290	0.00
37.	HKOO 299	1.30
38.	IPC 305	3.70
39.	MPJGK 7	0.00
40.	MPJGK 11	1.35
41.	GJG 2018	1.75
42.	GNG 9905	6.86
43.	IPC 2K25	12.60
44.	IPC 98-22	2.50
45.	PG 9425-5	1.55
46.	PG 9425-9	0.00
47.	BG 372	3.81
48.	BG 1103	7.60
49.	BGM 547	1.55
50.	GNG 1488	1.80
51.	H 82-2	3.02
52.	PBG 1	5.30
53.	PG 9425-9	2.13
54.	IPC 98-12	4.25
55.	RSG 963	20.15
56.	CSJ 104	3.65

S.No.	Entries	Root rot (%) Mean
57.	IPC 97-67	13.55
58.	RSG 143-1	3.15
59.	RSG 888	31.23
60.	RSG 7-931	9.50
61.	PG 9425-5	0.85
62.	BG 1053	2.64
63.	BG 1108	1.85
64.	BG 2024	1.20
65.	GNG 1499	23.65
66.	L 550	14.25
67.	MPJGK 6	9.80
68.	PBGK 220	17.20
69.	PG 97313	5.15
70.	PG 95333	4.05
71.	BGD 128	1.90
72.	MPJGK 4	0.00
73.	MPJGK 99-115	0.75
74.	MPJGK 3	0.00
75.	CSJ 140	18.10
76.	CSJ 160	11.03
77.	HHOO 38	0.00
78.	HHOO 53	0.00
79.	Phule G 96006	0.80
C	L 550 (Average)	27.62

DISCUSSIONS

CHAPTER - V

Discussion

Chickpea (*Cicer arietinum* L.) is one of the important pulse crop of India, accounting for more than one third of area and about 40 per cent of the production of pulses in the country. Amongst the biotic stresses root rot caused by *Macrophomina phaseolina* (Tassi) Goid [*Rhizoctonia bataticola* (Taub) Butler] is of considerable importance. Survey of root rot in chickpea around Sehore district indicated that the disease is prevalent in the region maximum upto 7.00 per cent at flowering and podding stage, causing death of the plant. The disease had also been reported from other parts of the State, particularly from northern Madhya Pradesh, ranging from 3.58 to 20.63 per cent (Gupta *et al.*, 1983). This disease is also of significance in other parts of the country and world (Reddy *et al.*, 1991 ; Aung *et al.*, 1992 and Beniwal *et al.*, 1992).

The characterization and identification of isolates of *M. phaseolina* is usually based on morphological criteria. In the present studies the fungus was identified as *Macrophomina phaseolina* based on standard mycological keys and pathogenicity tests on chickpea. The mycelium of the fungus was inter and intra cellular and formed numerous sclerotial bodies on the host tissues. The pycnidial stage was later produced on the host.

The dry root rot symptoms observed were mainly brittleness of tap root and presence of minute sclerotia. These symptoms separated this disease from the so called wilt complex "as termed by various workers in India. (Nene, 1989).

Substantial mortality and loss in chickpea due to *M. phaseolina* had been reported in high ambient temperature 30°C and above in the post - flowering stage (Singh and Mehrotra, 1982). Such situation is common in central and southern India. Vertisol supports the disease development whereas alfisol do not. The studies conducted on weather and edaphic factors indicated that maximum atmospheric temperature upto 34.5°C favoured the disease

development giving positive significant relationship. The same was true with increasing minimum atmospheric temperature. The disease incidence was found to decrease with increase in relative humidity in the atmosphere. Being a soil - born disease these finding required confirmation with edaphic parameters like soil temperature and soil moisture. As obvious the weather parameters also influenced these edaphic factors and hence the root rot development in infested soil. It is evident from the present studies that disease increased with the rise in soil temperature, while decreased with the higher soil moisture content. The studies may be helpful for the areas having irrigation facilities. Reducing soil moisture stress at post flowering can minimize the root rot incidence in chickpea.

Being soil borne disease which usually appear at post flowering stage in field, various efforts have been done to find out suitable technology for the management of this disease. Seed treatment with thiram + carbendazim @ 2 + 1 g/kg seed and with carboxin 2 g/kg were most effective in significantly reducing chickpea mortality in sick plot due to root rot, followed by thiram + carboxin. This revealed that systemic fungicides because of their long residual effects are more effective rather than non-systemic fungicide. A combination of both can also work very well in chickpea. Earlier studies of Verma (1976) also supports the present findings as seed treatment either with carbendazim or carboxin was effective upto 30 days. *T. viride* and *T. harzianum* are the potential antagonist against *M. Phaseolina* (Saxena & Saxena, 1995 ; Singh *et al.*, 1995). In the present study efforts have been made to find out compatible strain of *Trichoderma* spp. with carboxin. However, *Trichoderma harzianum* a strain from Project Directorate of Biological Control (PDBC) strain, Bangalore was equally compatible with carboxin as *T. harzianum* strain of Kanpur. This indicated that there is no difference in the strain of *T. harzianum* obtained from the two different places. It is also known from the present studies that *T. harzianum* reduced the root rot to significant levels as compared to *T. viride* (*Gulburga* str.) however, effectivity of *T. viride* @ logcfu/m against *M. phaseolina* in chickpea had been reported under pot condition (Shaid *et al.*, 2000).

The antagonistic activity in *T. viride* is due to the production of enzymes such as chitinases and β 1, 3, glucose (Mukhopadhyay, 1987).

The large scale application of bioagents particularly for the management of soil borne disease of chickpea is very much limited. This is because of non availability of suitable and cheaper substrate. Therefore, different substrates were used to multiply *T. viride* and thereafter its application in field before sowing of chickpea. It is evident from the findings that vermi compost and spent mushroom substrate (SMS) are suitable substrates rather than *T. viride* multiplied on field soil. This is because that vermi compost contains essential nutrients required by *T. viride* and its moisture retention capacity is high as compared to the field soil.

In SMS, mushroom mycelium is abundantly available and provides substrate for parasitization of bioagents. Moreover, SMS was also found effective in reducing inoculum of soil borne pathogen *Fusarium* in tomato and the disease index was reduced upto 67 - 74 per cent (Harender and Kapoor 1997) the low root rot incidence in chickpea may be either due to the buildup population of *T. viride* or due to the indirect effect of SMS as it improves the root quality and plant health (Cigl *et al.*, 1994). Significant reduction in the root rot incidence in all the treatments suggested that population of bioagents should be buildup in the rhizosphere, so that the pathogen inoculum is restricted. However, seed treatment with the bioagent reduced the root rot but not to the extent as it was controlled by the soil application. This may be due to the less buildup of *T. viride* population in rhizosphere though the highest cost benefit ratio was observed in seed treatment.

The experiment on efficacy of fungicides with bio agent was conducted to compare both the treatments. The effect of seed treatment of *T. viride* either alone or in combination with carboxin was equally effective in reducing the root rot incidence. This may be due to long effectivity of then bio agent as the disease appeared at post flowering stage of the crop till then the effect of the fungicidal

seed treatment is reduced and on the other hand population of *T. viride* buildup in rhizosphere, protect the plant from *M. phaseolina* infection. Spray on collar region of the crop was not effective at one month old crop. This is because of the fact that the pathogen had already established a pathogenic relationship with the host. The cost benefit ratio was as good as 1 : 80, at par with the seed treatment with carbendazim + thiram.

Cropping system is the most important component in integrated disease management practices. Intercropping of linseed with chickpea alone could not reduce root rot incidence as compared to sole chickpea crop, but when other approaches were included the root rot incidence was reduced. Hence, the finding emphasized the importance of integrated management practices for minimizing the root rot incidence in chickpea as no single method is dependable for controlling the soil borne disease like chickpea root rot.

SUMMARY & CONCLUSION

CHAPTER VI

Summary and Conclusion

Root rot of chickpea caused by *Macrophomina phaseolina* is regular in occurrence and its severity is increasing in Madhya Pradesh. Due to its wide host range it is difficult to control this soil borne disease, present investigation was taken up to study relationship of root rot with meteorological parameters (both climatic and edaphic), efficacy of different seed treating agents chemical as well as biological, different substrates for mass scale multiplication of *Trichoderma viride* and their effect on root rot incidence, methods of application of fungicides and bio agents and integration of different components for the management of root rot. The salient findings of these studies are as under

1. The disease appeared at post-flowering stage of chickpea and prevalent in almost all the places surveyed in Sehore district. The incidence is more in old varieties, while improved varieties are less affected. The disease exhibited characteristic symptoms of root rot, drooping of petioles and leaflets confined to very top of the plant. The leaves and stem of affected plants are straw coloured tap root is dark, devoid of lateral and finer roots, presence of dark minute sclerotial bodies in the roots.

2. The root rot incidence was influenced by both environmental as well as edaphic factors. The maximum temperature had a positive relationship, the disease increased 0.63% with the rise in 1°C maximum temperature. Similarly, the root rot increased 0.83% with the 1°C increase in minimum temperature. The relative humidity had a negative relationship the disease incidence decreased by 0.41% with the 1% increase in relative humidity. Soil temperature exhibited greater role than the other parameters. The disease incidence increased by 1.14% with 1°C increase in soil temperature.

3. Under field conditions seed treatment with Kanpur and PDBC strains of *Trichoderma harzianum* and carboxin were found to be comparable and effective in reducing root rot incidence. However, seed treatment with carboxin and thiram + carbendazim were most effective. The latter gave maximum yield of chickpea and net returns.

4. *Trichoderma viride* multiplied on vermi compost and mushroom substrate (SMS) were most effective in reducing root rot incidence, as compared to the other substrates like farm yard manure, nadep compost and field soil. The maximum net returns were also noted in these treatments.

5. Seed treatment with either carbendazim + thiram or *T. viride* was effective in significantly reducing root rot incidence in chickpea. However spray of carbendazim on collar region of one and half month old crop could also reduce the root rot incidence upto significant levels, as compared to control. The maximum yield and per hectare net returns were in seed treatment with carbendazim + thiram.

6. Under IDM approach seed treatment with the combination of thiram + carbendazim (2 : 1) and intercropping of chickpea with linseed (4 : 2) had higher chickpea equivalent yield in comparison to sole chickpea crop. The root rot incidence was also least in dual practice.

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APPENDICES

APPENDICES

Experiment A : The Skeleton of analysis of variable table.

S.No.	Source of Variance	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Block	r - 1				
2.	Treatments	t - 1				
3.	Error	(r-1) (t-1)				
4.	Total					

The difference between two means was subjected to further testing by computing critical difference at 5 % probability level

The following formula used by various estimations for

1. Standard error for treatment means.

S. Em. = ± √Ve/r

2. Critical difference = C. D. = S. Em. x √2 x 't'

Value of 5% probability level at error degrees of freedom.

Where,

Ve = Error mean sum of square

R = Number of replication

t = t value of 5% probability level

S.Em. = Standard error of means

C.D. = Critical difference

Anova for Table 6

Analysis of variance for root rot incidence

S.No.	Source of Variance	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	2	10.59	5.29	12.41	2.31
2.	Treatments	11	124.37	11.30		
3.	Error	22	20.03	0.91		
4.	Total	35		S.Em ± 0.55 CD at 5% 1.61		

Anova for Table 6

Analysis of variance for the yield

S.No.	Source of Variance	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	2	1901.6	950.8	5.22	2.31
2.	Treatments	11	95398.2	8672.56		
3.	Error	22	36510.56	1659.57		
4.	Total	35	S.Em ± 23.52 CD at 5% 68.97			

Anova for Table 8

Analysis of variance for root rot incidence

S.No.	Source of Variance	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	2	19.42	9.71	3.40	2.78
2.	Treatments	7	72.48	10.35		
3.	Error	14	42.53	3.03		
4.	Total	23		S.Em ± 1.23 CD at 5% 2.64		

Anova for Table 8

Analysis of variance for the yield

S.No.	Source of Variance	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	2	10086.21	5043.10	3.86	2.78
2.	Treatments	7	773146.30	110449.47		
3.	Error	14	401102.54	286.50.18		
4.	Total	23	S.Em ± 91.40 CD at 5% 277.25			

Anova for Table 10

Analysis of variance for root rot incidence

S.No.	Source of Variance	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	2	17.25	8.62	26.09	2.78
2.	Treatments	7	343.29	49.04		
3.	Error	14	26.30	1.87		
4.	Total	23		S.Em ± 0.69 CD at 5% 2.08		

Anova for Table 10

Analysis of variance for the yield

S.No.	Source of Variance	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	2	31025.30	15512.65	3.32	2.78
2.	Treatments	7	616461.67	88065.95		
3.	Error	14	371687.32	26549.09		
4.	Total	23	S.Em $\pm$ 81.47 CD at 5% 247.14			

Anova for Table 12

Analysis of variance for the IDM on root rot incidence

S.No.	Source of variation	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	3	10.51	3.50	5.64	3.29
2.	Main	1	0.18	0.18	0.30	4.54
3.	Sub	2	83.74	41.87	69.96	3.68
4.	Interation	2	1.68	0.84	1.40	3.68
5.	Error	15	8.99	0.60		
		Main	Sub	INT		
	S.Em ±	0.22	0.27	0.39		
	C.D. at 5%	NS	0.84	NS		

Anova for Table 12

Analysis of variance for the IDM on chickpea yield

S.No.	Source of variation	D. F.	S.S.	M.S.S	Cal. F.	Tab. F.
1.	Replication	3	41650.24	13883.41	4.31	3.29
2.	Main	1	742185.48	742185.48	230.29	4.54
3.	Sub	2	333292.38	166648.19	51.71	3.68
4.	Interation	2	34100.27	17050.13	2.29	3.68
5.	Error	15	18313.18	3222.90		
		Main	Sub	INT		
	S.Em ±	16.39	20.07	28.39		
	C.D. at 5%	NS	61.84	NS		

VITA

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